

ECO 209 Assignment 3

Name: _____

1. (10 pts) Use the table below

Period	Current Output	Potential Output	Gap (%)
1	5,500	5,000	a
2	b	5,500	3.64%
3	5,800	c	-4.92%
4	d	6,800	-0.74%
5	8,050	e	3.21%

Solve a) to e).

Solution

$$(a) \frac{5500}{5000} - 1 = 0.1$$

$$(b) (1 + 0.0364) \times 5500 = 5700$$

$$(c) \frac{5800}{1-0.0492} = 6100$$

$$(d) (1 - 0.0074) \times 6800 = 6749.68$$

$$(e) \frac{8050}{1+0.0321} = 7800$$

2. (20 pts) This question centers on the IS model from Chapter 11. In a closed economy, the consumption (C_t), investment (I_t), and government purchases (G_t) are represented by the following equations:

$$C_t = \bar{a}_c \tilde{Y}_t$$

$$I_t = \bar{a}_I \tilde{Y}_t - \bar{b} (R_t - \bar{r}) \tilde{Y}_t$$

$$G_t = \bar{a}_G \tilde{Y}_t + \tilde{\beta} \tilde{Y}_t \tilde{Y}_t$$

The equations for C and I remain the same as in the simple IS model, whereas the G equation incorporates an extension to account for automatic stabilizers – policy that automatically adjust to economic conditions to stabilize output, such as progressive taxes and unemployment benefits.

- (a) (3 pts) Considering the extension term to account for automatic stabilizers, what is the expected range for the parameter $\tilde{\beta}$?
- (b) (4 pts) Derive the IS curve.
- (c) (3 pts) How does the new extension $\tilde{\beta} \tilde{Y}$ alter the IS curve?
- (d) (4 pts) The economy is facing a recession of $\tilde{Y} = 0.05$. The government decided to implement a discretionary policy. By how much does the government need to increase the $\bar{a}_G$ parameter?
- (e) (2 pts) Do automatic stabilizers support the government's discretionary policy efforts (as in part d)) to stimulate GDP?
- (f) (4 pts) Use a graph to illustrate the effect of an increase in $\tilde{\beta}$. Draw two IS curves on a graph, with the real interest rate on the Y-axis and short-run output ($\tilde{Y}$) on the X-axis. Label IS_2 as the IS curve with a higher absolute value of $\tilde{\beta}$ compared to IS_1 .

Solution

- (a) The parameter $\bar{\beta}$ would be negative to capture the idea of automatic stabilizer. As the short-run output increase (fall), the government would reduce (increase) its spending.
- (b) Either of the following would be accepted

$$Y_t = C_t + I_t + G_t = \bar{a}_c \bar{Y}_t + \bar{a}_I \bar{Y}_t - \bar{b} (R_t - \bar{r}) \bar{Y}_t + \bar{a}_G \bar{Y}_t + \bar{\beta} \tilde{Y}_t \bar{Y}_t$$

$$\frac{Y_t}{\bar{Y}_t} = \frac{(\bar{a}_c + \bar{a}_I + \bar{a}_G) \bar{Y}_t - \bar{b} (R_t - \bar{r}) \bar{Y}_t + \bar{\beta} \tilde{Y}_t \bar{Y}_t}{\bar{Y}_t}$$

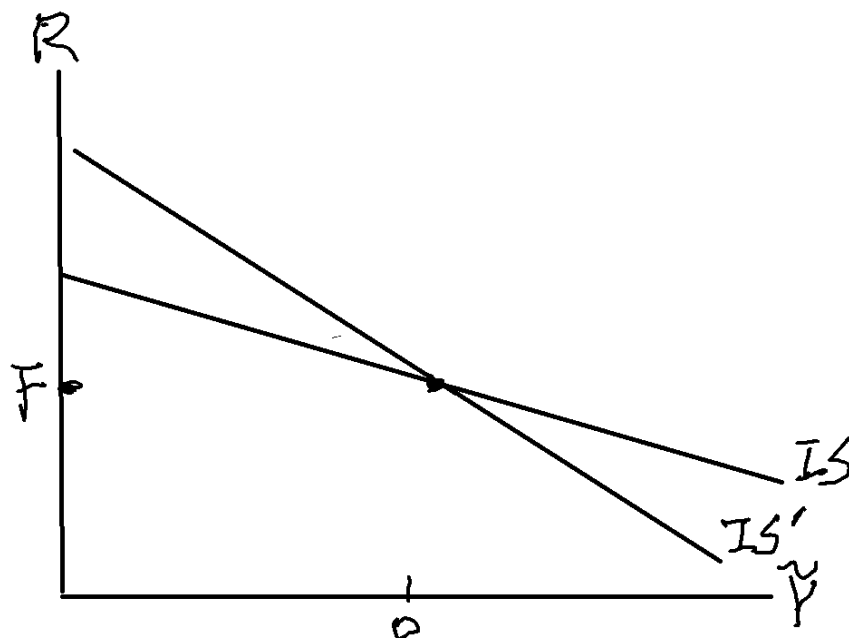
$$\frac{Y_t}{\bar{Y}_t} - 1 = \tilde{Y} = (\bar{a}_c + \bar{a}_I + \bar{a}_G - 1) - \bar{b} (R_t - \bar{r}) + \bar{\beta} \tilde{Y}_t$$

$$\tilde{Y}_t - \bar{\beta} \tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

$$\tilde{Y}_t = \frac{1}{1 - \bar{\beta}} \left[\bar{a} - \bar{b} (R_t - \bar{r}) \right]$$

- (c) In this setup, extension would work like a multiplier but in a opposite effect. It is because the parameter $\bar{\beta}$ is negative. Therefore, for one percent increase in, say, $\bar{a}$ would lead to a less than one percent increase in the short-run output.
- (d) The government need to increase $\bar{a}_G$ by $5\% \times (1 - \bar{\beta})$.
- (e) Given $\bar{\beta} < 0$, the government would need to increase government spending by more than the fall of short-run output

- (f) As the absolute value of β increases, the slope of the IS curve would be lower. The multiplier effect becomes smaller.



3. (15 pts) In a hypothetical economy, Brossard, the output gap fall to -8% during the Great Recession in 2009. Many people believe that the multiplier was equal to 2.5. Given the equations that describe the economy:

$$C_t = \bar{a}_c \bar{Y}_t + \bar{x} \tilde{Y}_t \bar{Y}$$

$$I_t = \bar{a}_I \bar{Y}_t - \bar{b} (R_t - \bar{r}) \bar{Y}_t$$

$$G_t = \bar{a}_G \bar{Y}_t$$

The equations for G and I remain the same as in the simple IS model, whereas the C equation incorporates an extension to account for the imperfect consumption smoothing.

- (a) (4 pts) Derive the IS curve.
- (b) (2 pts) What is value of $\bar{x}$?
- (c) (2 pts) What is value of $\bar{x}$ if the multiplier was 1.25?
- (d) (4 pts) What would the percent change in government expenditure be to close this gap, assuming monetary policy is not being used and the multiplier is equal to 2?
- (e) (3 pts) If there is no aggregate shock (i.e., $\bar{a} = \bar{a}_c + \bar{a}_I + \bar{a}_G - 1 = 0$), $\bar{b} = 0.5$, $\bar{r} = 0.05$, what would the real interest rate R_t be?

Solution

(a)

$$C_t + I_t + G_t = Y_t = \bar{a}_c \bar{Y}_t + \bar{x} \tilde{Y}_t + \bar{a}_I \bar{Y}_t - \bar{b} (R_t - \bar{r}) \bar{Y}_t + \bar{a}_G \bar{Y}_t$$

$$\frac{Y_t}{\bar{Y}_t} = \bar{a}_c + \bar{x} \tilde{Y}_t + \bar{a}_I - \bar{b} (R_t - \bar{r}) + \bar{a}_G$$

$$\frac{Y_t}{\bar{Y}_t} - 1 = \tilde{Y}_t = [\bar{a}_c + \bar{a}_I + \bar{a}_G - 1] + \bar{x} \tilde{Y}_t - \bar{b} (R_t - \bar{r})$$

$$\tilde{Y}_t - \bar{x} \tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

$$\tilde{Y}_t = \frac{1}{1 - \bar{x}} [\bar{a} - \bar{b} (R_t - \bar{r})]$$

(b) The multiplier is $\frac{1}{1-\bar{x}}$. If the multiplier is equal to 2.5,

$$2.5 = \frac{1}{1 - \bar{x}}$$

$$\bar{x} = 0.6$$

(c) The $\bar{x} = 0.2$

(d) The government has to increase its expenditure by $8\%/2 = 4\%$

(e)

$$-0.08 = 2.5[0 - 0.5(R_t - 0.05)]$$

$$-0.08 = 2.5[0 - 0.5(R_t - 0.05)]$$

$$R_t = 0.114$$

*Since this part of the question didn't specify the multiplier, one may use any of the three numbers used above.

4. (10 pts) The quantity equation states that

$$M_t V_t = P_t Y_t$$

- (a) (3 pts) What assumptions about M_t , V_t , and Y_t are necessary for the quantity theory of money to consider the money supply as the key determinant of the price level in the long run?
- (b) (2 pts) Show that the inflation rate is a function of *only* the growth rates of money supply (g_M) and output (g_Y). State all the necessary assumptions.
- (c) (3 pts) Suppose the central bank announces it will gradually slow down the money supply growth. What are the likely effects on the inflation rate initially and in the long run?
- (d) (2 pts) In the long run, will the price level be higher or lower? What about the output?

Solution

(a) M_t is controlled by the central and is exogenously given

V_t is constant in the long run

Y_t does not affected by the money market and is determined by only the real variables, according to the classical dichotomy

Thus, P_t is determined by the M_t only

(b)

$$g_m + g_v = g_P + g_Y$$

Let $g_v = 0$ as it is usually assume to be a constant. Therefore,

$$g_P = \pi = g_m - g_Y$$

(c) The inflation rate would slowly decrease. In the long run, the inflation would be lower.

(d) The price level would be higher than before, even though inflation is lower, as prices would still rise but at a slower rate. Output, however, would remain unchanged, as it is determined by factors unrelated to nominal variables.

5. (35 pts) Data Question:

You may use any program of your choice to complete the following question.

(a) (4 pts) Download the following data series from FRED:

- "Real Gross Domestic Product for Canada [NGDPRSAXDCCAQ]" covering the period from 1961Q1 to 2024Q2.

(b) (6 pts) Compute the 5-quarter trailing moving average of Real GDP for Canada. Use this as the GDP trend and calculate the cyclical component (short-run output) as a percentage deviation from the trend.

(c) (8 pts) Following Hamilton's approach, estimate the regression model given by:

$$y_{t+8} = 12614.50418 + 0.54737y_t + 0.05201y_{t-1} + 0.14761y_{t-2} + 0.27181y_{t-3} + v_{t+8}$$

Using the estimated model, calculate the trend and cyclical component (as a percentage deviation from the trend) of Real GDP for Canada.

(d) (8 pts) Plot the two cyclical components obtained in part (b) and part (c) on a single graph with:

- **Y-axis:** Cyclical component values from part b) and part c).
- **X-axis:** Time (display only the year and month, excluding the day).
- **Titles:** Provide appropriate titles for the Y-axis, X-axis, and the overall chart.

(e) (8 pts) Analyze the economic impact during the recent pandemic crisis. Based on the graph from part (d), which method provides a clearer picture of the economy? (in 5-10 sentences)

(This is an open-ended question. Your response will be evaluated based on how well you interpret the graph and your understanding of the Moving Average and Hamilton Filter methods.)

*R Code for Estimating the Model in Part (c)

```
#Clear the Environment

rm(list = ls())

#Loading Package

install.packages("neverhpfiler")

library(neverhpfiler)

library(xts)      # For time series handling


#Loading Data

# Replace ... with actual file path to the Data
GDP_data <- read.csv(".../NGDPRSAXDCCAQ.csv")
GDP_data[[1]] <- as.Date(GDP_data[[1]])
GDP <- xts(GDP_data[[2]], order.by = GDP_data[[1]])
colnames(GDP) <- "GDP_Value"


#Estimate Hamilton filter

gdp_model <- yth_glm(GDP, h = 8, p = 4)

summary(gdp_model)
```

– The End –